

Janamaitri Multiple Campus

(Tribhuvan University)
Kuleshwor , Kathmandu



Lab Report of Probability & Statistics (CAST-202)

Faculty of Humanities & Social Sciences

Tribhuvan University

Kuleshwor, Nepal

Submitted By

Name: Abishek Gharti

Magar

Roll No:

Submitted To

Janamaitri Multiple Campus

Kuleshwor, Nepal



जनमैत्री बहुमुखी क्याम्पस

त्रिभुवन विश्वविद्यालयबाट सम्बन्धन प्राप्त

विश्वविद्यालय अनुदान आयोगबाट गुणस्तर प्रत्यायनकृत (२०७६)

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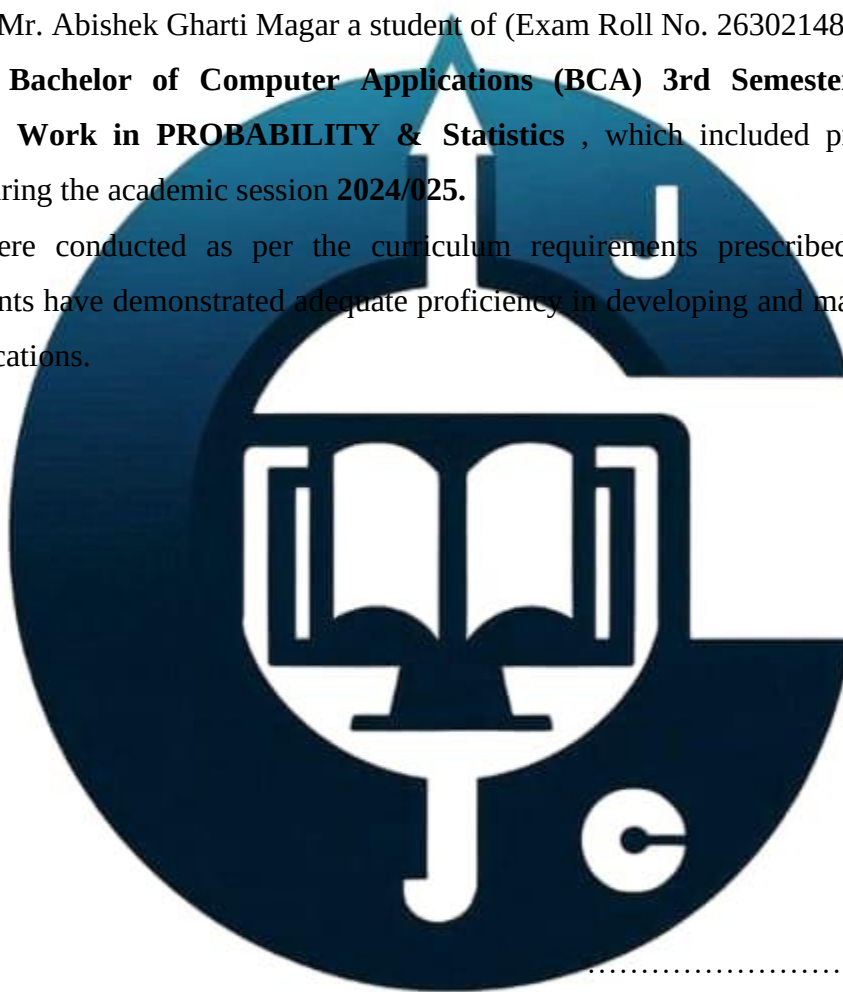
च.नं. (Ref. No)

मिति: 2082-04-13

Letter of Certificate

This is to certify that Mr. Abishek Gharti Magar a student of (Exam Roll No. 26302148 and TU Registration No.) **Bachelor of Computer Applications (BCA) 3rd Semester** have successfully completed their **Lab Work in PROBABILITY & Statistics** , which included practical training and exercises in **SPSS**, during the academic session **2024/025**.

The lab sessions were conducted as per the curriculum requirements prescribed by the Tribhuvan University, and students have demonstrated adequate proficiency in developing and managing dynamic and responsive web applications.



.....
Mr. Kamal Tamrakar
(Head of Department)



जनमैत्री बहुमुखी क्याम्पस

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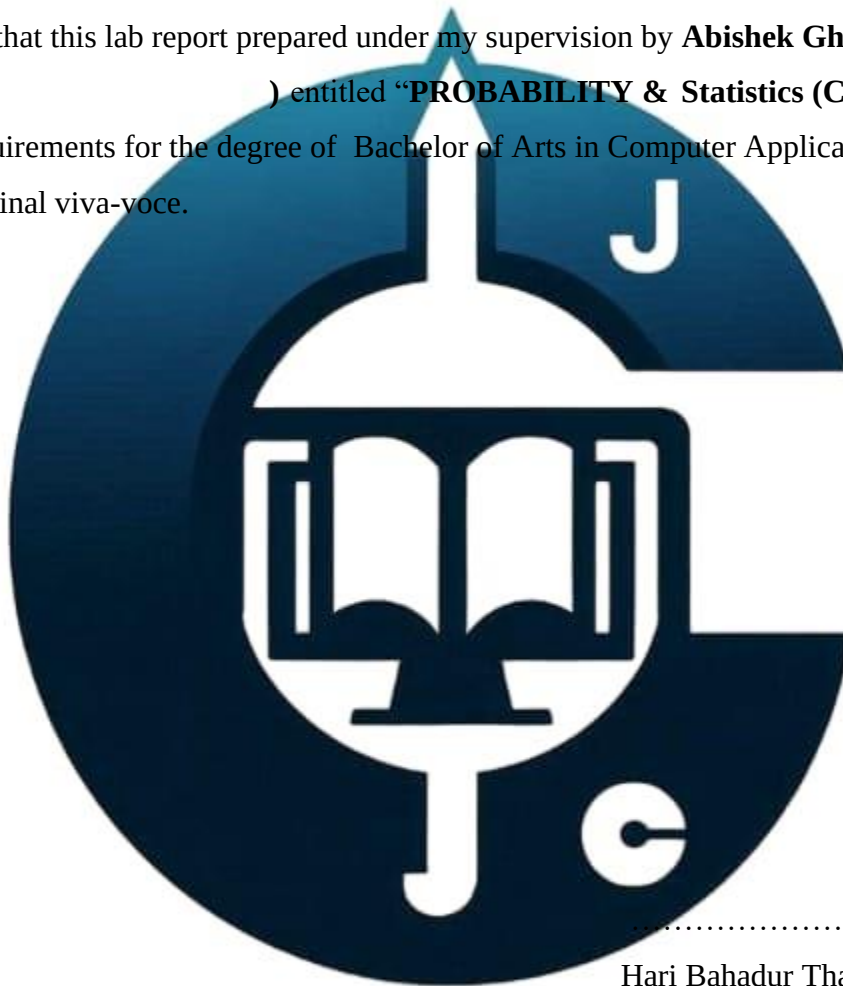
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मिति: 2082-04-13

Recommendation for Acceptance

I hereby recommend that this lab report prepared under my supervision by **Abishek Gharti Magar**

(Redg No:) entitled “**PROBABILITY & Statistics (CAST-202)**” in partial fulfillment of the requirements for the degree of Bachelor of Arts in Computer Application is recommended for the evaluation in final viva-voce.



.....
Hari Bahadur Thapa (Supervisor)

Date:- 2082/04/13



जनमैत्री बहुमुखी क्याम्पस

त्रिभुवन विश्वविद्यालयबाट सम्बन्धन प्राप्त

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च.नं. (Ref. No)

मिति: 2082-04-13

Letter of Approval

This Lab Report entitled “**PROBABILITY & Statistics (CAST-202)**” submitted by Mr. Abishek Gharti Magar in partial fulfillment of the requirements for the bachelor degree Bachelor of in Computer Application for the Web Technology course of BCA 3rd Semester has been approved.

Viva Voce Committee

Signature

Mr. Kamal Tamrakar

(Head of Department)

Mr. Rohit Poudel

(External Supervisor)

Mr. Hari Bahadur Thapa

(Internal Supervisor)

Date :- 2082/04/13

Table of Contents

ACKNOWLEDGEMENT	1
PREFACE	2
1. SPSS in Statistical Computations.....	1
2. How SPSS Helps in Research & Data Analysis Programs?	3
i. Data Transformation.....	3
ii. Regression Analysis	3
iii. Analysis of Variance (ANOVA)	3
iv. Multivariate analysis of variance (MANOVA).....	3
v. T-tests.....	3
Solved Questions	4
3. Enter the following data in the entry sheet of SPSS. Prepare data entry sheet in SPSS to enter the above variables (name, roll_no, age, weight (in kg), sex).....	4
4. Calculate mean, median, mode, standard deviation, variance, range, minimum, maximum value, skewness and kurtosis of the following data.....	5
5. Enter the following values in SPSS and find the regression equation of y on x.....	7
6. Enter the following values in SPSS and find the correlation between X.....	8
7. The weight of a number of copper wires each of length 1m were obtained. These are shown below classified according to die from which they come	9
8. Are the weights of copper wire for different die are equal?	10
9. From the following grouped data find mean, median, quartiles, standard deviation and variance.	11

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At the very beginning, I would like to express my deepest gratitude to all the staffs and teachers for giving me the strength and the composure to complete my BCA third semester course and prepare this report within the scheduled time.

During the period of my internship work, I have received generous help from many quarters, which I like to put on record here with deep gratitude and great pleasure.

First and foremost, I am grateful to my supervisor, Mr. Hari Bahadur Thapa sir. He allowed me to encroach upon his precious time freely right from the very beginning of this research work until the completion of my report.

His guidance, encouragement, and suggestions provided me the necessary insight into the research problem and paved the way for the meaningful ending of the work in a short duration.

I have no hesitation to say that, without his constant support and valuable advice from time-to-time, I would probably fail to complete the work in an appropriate manner.

My special gratitude goes to Kamal Tamrakar (Director of BCA program) for his modern outlook and meticulous supervision to carry out the job perfectly.

I am also thankful to my friends, my parents and all the family members for their sincere co-operation and supporting me for making a standard report.

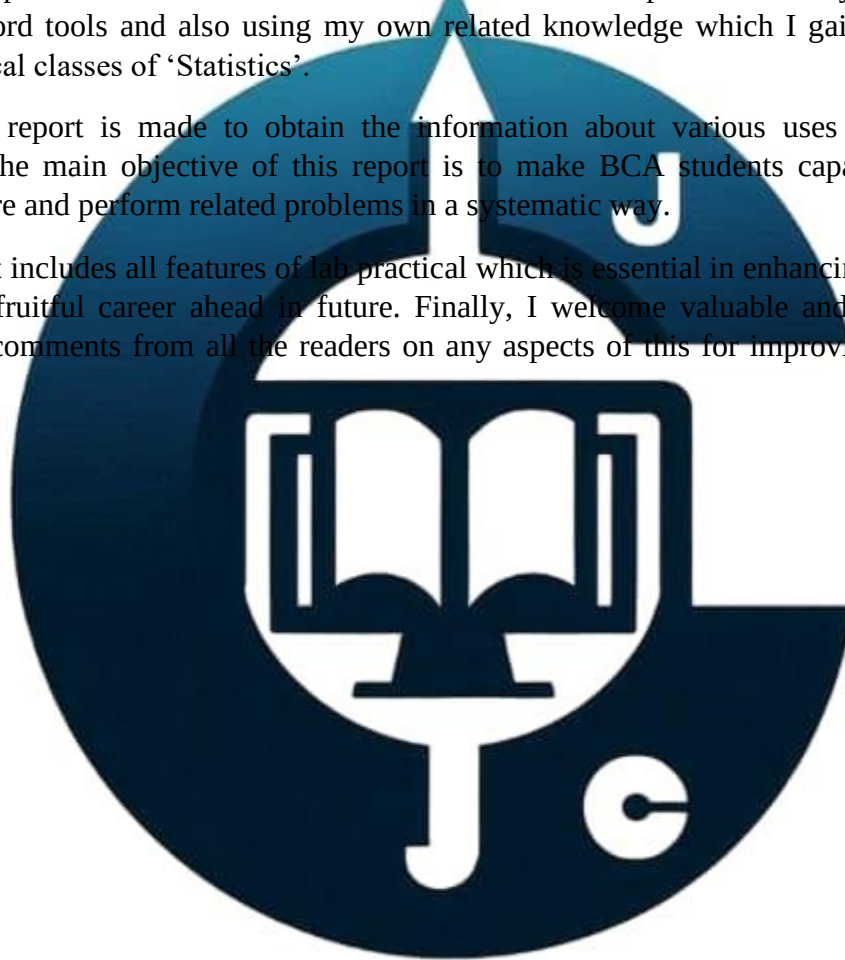
PREFACE

SPSS is a program widely used for statistical analysis in the social science it is also used in market researcher medical researcher government and so many places. It was originally created by three graduate students at Stanford university in the late 1960s..

This lab report is assigned to students as they lag behind in fair result processing and to solve the current problem of various academic institutions. The report is made by using SPSS software, MS Word tools and also using my own related knowledge which I gained from the theory and practical classes of 'Statistics'.

This Lab report is made to obtain the information about various uses of statistical software tools. The main objective of this report is to make BCA students capable of using Statistical software and perform related problems in a systematic way.

The report includes all features of lab practical which is essential in enhancing the skill of students for the fruitful career ahead in future. Finally, I welcome valuable and constructive suggestions and comments from all the readers on any aspects of this for improving my logic.



1. SPSS in Statistical Computations

The statistical software namely Statistical Package for the Social Sciences shortly abbreviated as SPSS was originally created by three graduate level students of Stanford University towards late 1960s. Later, IBM Company purchased SPSS in 2009, and then the name of the SPSS had been changed into IBM SPSS Statistics or in generally speaking language, simply IBM SPSS. This software has become one of the most popular and successful statistical software among may who involve in the data analysis in different areas of studies. Even today to the users of this software generally used to call it as SPSS not IBM SPSS because of popularity of the name SPSS among the wide users.

SPSS is a widely used program for statistical analysis in social science. It is also used by market researchers, health researchers, survey companies, government, education researchers, marketing organizations, data miners, and others. The original SPSS manual (Nie, Bent & Hull, 1970) has been described as one of "sociology's most influential books" for allowing ordinary researchers to do their own statistical analysis. In addition to statistical analysis, data management (case selection, file reshaping, creating derived data) and data documentation (a metadata dictionary is stored in the data file) are features of the base software.

SPSS Statistics places constraints on internal file structure, data types, data processing, and matching files, which together considerably simplify programming. SPSS datasets have a two-dimensional table structure, where the rows typically represent cases (such as individuals or households) and the columns represent measurements (such as age, sex, or household income). Only two data types are defined: numeric and text (or "string"). All data processing occurs sequentially case-by-case through the file (dataset). Files can be matched one-to-one and one-to-many, but not many-to-many. In addition to that cases-by-variables structure and processing, there is a separate Matrix session where one can process data as matrices using matrix and linear algebra operations.

The many features of SPSS Statistics are accessible via pull-down menus or can be programmed with a proprietary 4GL *command syntax language*. Command syntax programming has the benefits of reproducible output, simplifying repetitive tasks, and handling complex data manipulations and analyses. Additionally, some complex applications can only be programmed in syntax and are not accessible through the menu structure. The pull-down menu interface also generates command syntax: this can be displayed in the output, although the default settings have to be changed to make the syntax visible to the user. They can also be pasted into a syntax file using the "paste" button present in each menu. Programs can be run interactively or unattended, using the supplied Production Job Facility.

Additionally, a "macro" language can be used to write command language subroutines. A Python programmability extension can access the information in the data dictionary and data and dynamically build command syntax programs. The Python programmability extension, introduced in SPSS 14, replaced the less functional SAX Basic "scripts" for most purposes, although SaxBasic remains available. In addition, the Python extension allows SPSS to run any of the statistics in the free software package R. From version 14 onwards, SPSS can be driven

externally by a Python or a VB.NET program using supplied "plug-ins". (From Version 20 onwards, these two scripting facilities, as well as many scripts, are included on the installation media and are normally installed by default.)

The graphical user interface has two views which can be toggled by clicking on one of the two tabs in the bottom left of the SPSS Statistics window. The 'Data View' shows a spreadsheet view of the cases (rows) and variables (columns). Unlike spreadsheets, the data cells can only contain numbers or text, and formulas cannot be stored in these cells. The 'Variable View' displays the metadata dictionary where each row represents a variable and shows the variable name, variable label, value label(s), print width, measurement type, and a variety of other characteristics. Cells in both views can be manually edited, defining the file structure and allowing data entry without using command syntax. This may be sufficient for small datasets. Larger datasets such as statistical surveys are more often created in data entry software, or entered during computer-assisted personal interviewing, by scanning and using optical character recognition and optical mark recognition software, or by direct capture from online questionnaires. These datasets are then read into SPSS.

Statistical output is to a proprietary file format (*.spv file, supporting pivot tables) for which, in addition to the in-package viewer, a stand-alone reader can be downloaded. The proprietary output can be exported to text or Microsoft Word, PDF, Excel, and other formats. Alternatively, output can be captured as data (using the OMS command), as text, tab-delimited text, PDF, XLS, HTML, XML, SPSS dataset or a variety of graphic image formats (JPEG, PNG, BMP, and EMF).

2. How SPSS Helps in Research & Data Analysis Programs?

i. **Data Transformation**

This technique is used to convert the format of the data. After changing the data type, it integrates same type of data in one place and it becomes easy to manage it. You can insert the different kind of data to SPSS and it will change its structure as per the system specification and requirement. It means that even if you change the operating system, SPSS can still work on old data.

ii. **Regression Analysis**

It is used to understand the relation between dependent and interdependent variable that are stored in a data file. It also explains how a change in the value of an interdependent variable can affect the dependent data. The primary need of regression analysis is to understand the type of relationship between different variables.

iii. **Analysis of Variance (ANOVA)**

It is statistical approach to compare events, groups or processes and find out the different between them. It can help you understand which method is more suitable for executing a task. By looking at the result, you can find the feasibility and effectiveness of the particular method.

iv. **Multivariate analysis of variance (MANOVA)**

This method is used to compare data of random variables whose value is unknown. MANOVA technique can also be used to analyze different types of population and what factors can affect their choices.

v. **T-tests**

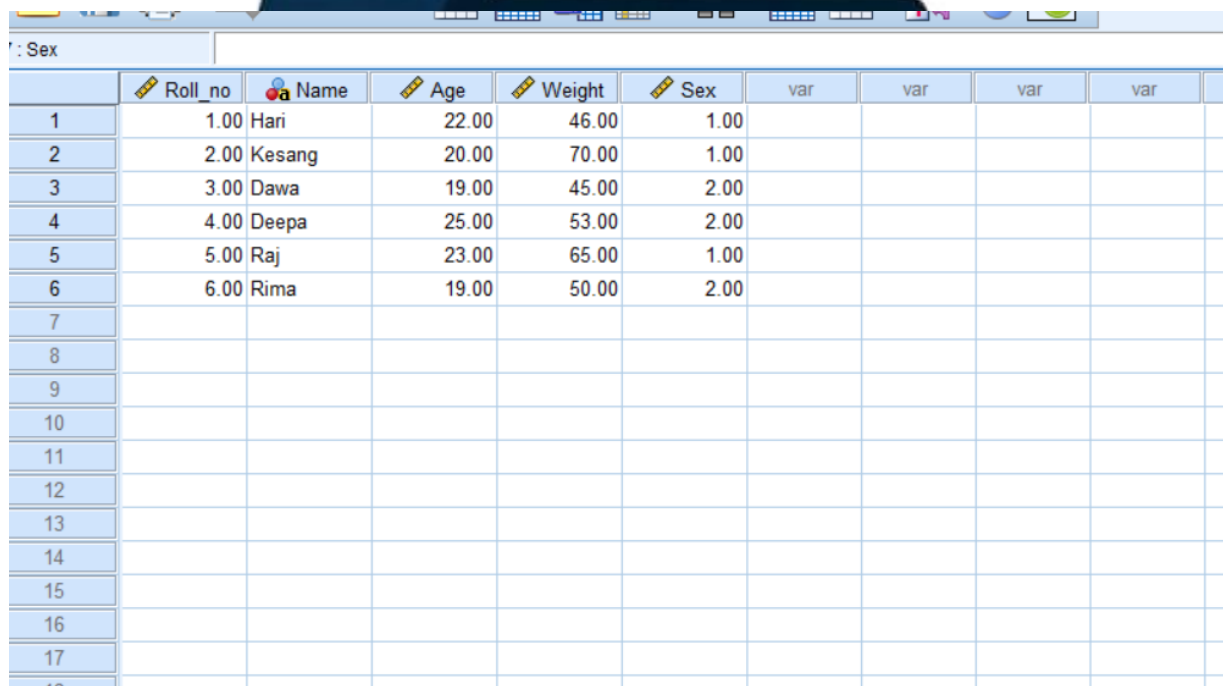
It is used to understand the difference between two sample types and researchers apply this method to find out the difference in the interest of two kinds of groups. This test can also understand if the produced output is meaningless or useful.

Solved Questions

3. Enter the following data in the entry sheet of SPSS. Prepare data entry sheet in SPSS to enter the above variables (name, roll_no, age, weight (in kg), sex).

Roll_no	Name	Age	Weight	Sex
1	Hari	22	46	1
2	Kesang	20	70	1
3	Dawa	19	45	2
4	Deepa	25	53	2
5	Raj	23	65	1
6	Rima	19	50	2

Solution:



The screenshot shows the SPSS Data Editor window. The 'Data View' tab is active, displaying a data grid with 18 rows and 10 columns. The first six rows contain the data from the table above, with values entered in the columns for Roll_no, Name, Age, Weight, and Sex. The remaining rows are empty. The column headers are Roll_no, Name, Age, Weight, Sex, and four empty columns labeled 'var'.

	Roll_no	Name	Age	Weight	Sex	var	var	var	var
1	1.00	Hari	22.00	46.00	1.00				
2	2.00	Kesang	20.00	70.00	1.00				
3	3.00	Dawa	19.00	45.00	2.00				
4	4.00	Deepa	25.00	53.00	2.00				
5	5.00	Raj	23.00	65.00	1.00				
6	6.00	Rima	19.00	50.00	2.00				
7									
8									
9									
10									
11									
12									
13									
14									
15									
16									
17									
18									

4. Calculate mean, median, mode, standard deviation, variance, range, minimum, maximum value, skewness and kurtosis of the following data.

Name	Age	Weight
Kim	20	80
Hari	25	65
Reeta	50	60
Reena	45	65
Shyam	25	55
Rohan	25	60
Deepa	25	65
Sarala	20	50
Sara	25	45
Rajesh	50	55

Solution:

File Edit View Data Transform Analyze Graphs Utilities Extensions Window Help											
10 : Weight											
	Name	Age	Weight	var	var	var	var	var	var	var	var
1	kim	20.00	80.00								
2	Hari	25.00	65.00								
3	Reeta	50.00	60.00								
4	Shyam	25.00	60.00								
5	Rohan	25.00	60.00								
6	Deepa	25.00	65.00								
7	Sarala	20.00	50.00								
8	Sara	25.00	45.00								
9	Rajesh	50.00	55.00								
10											
11											
12											
13											
14											
15											
16											
17											
18											
19											
20											

Frequencies

[DataSet1]

		Age	Weight
N	Valid	9	9
	Missing	0	0
Mean		29.4444	60.0000
Median		25.0000	60.0000
Mode		25.00	60.00
Std. Deviation		11.84389	10.00000
Variance		140.278	100.000
Skewness		1.475	.603
Std. Error of Skewness		.717	.717
Kurtosis		.532	1.388
Std. Error of Kurtosis		1.400	1.400
Range		30.00	35.00
Minimum		20.00	45.00
Maximum		50.00	80.00

Frequency Table

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	20.00	2	22.2	22.2	22.2
	25.00	5	55.6	55.6	77.8
	50.00	2	22.2	22.2	100.0
Total		9	100.0	100.0	

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	45.00	1	11.1	11.1	11.1
	50.00	1	11.1	11.1	22.2
	55.00	1	11.1	11.1	33.3
	60.00	3	33.3	33.3	66.7
	65.00	2	22.2	22.2	88.9
	80.00	1	11.1	11.1	100.0
Total		9	100.0	100.0	

5. Enter the following values in SPSS and find the regression equation of y on x

X	2	5	5	8	5	9	8
Y	5	7	2	4	6	2	6

Solution:

8 : y

	x	y	var	var	var	var	var	v
1	2.00	5.00						
2	5.00	7.00						
3	5.00	2.00						
4	8.00	4.00						
5	5.00	6.00						
6	9.00	2.00						
7	8.00	6.00						
8								
9								
10								
11								
12								
13								

➔ **Regression**

[DataSet2]

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	y ^b		Enter

a. Dependent Variable: x

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.308 ^a	.095	-.086	2.55282

a. Predictors: (Constant), y

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	3.416	1	3.416	.524	.502 ^b
	Residual	32.584	5	6.517		
	Total	36.000	6			

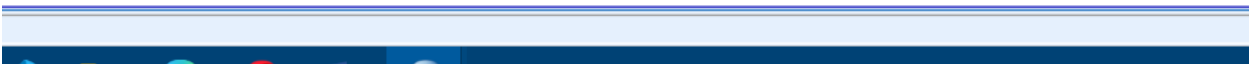
a. Dependent Variable: x

b. Predictors: (Constant), y



Coefficients ^a					
Model		Standardized Coefficients		Standardized Coefficients	Sig.
		B	Std. Error	Beta	
1	(Constant)	7.735	2.583		.030
	y	-.380	.524	-.308	.502

a. Dependent Variable: x



6. Enter the following values in SPSS and find the correlation between X

X	30	58	25	65	70	65
Y	20	50	45	65	45	60

7: y								
	x	y	var	var	var	var	var	
1	30.00	20.00						
2	58.00	50.00						
3	25.00	45.00						
4	65.00	65.00						
5	70.00	45.00						
6	65.00	60.00						
7								
8								
9								
10								
11								
12								
13								
14								
15								
16								

→ Correlations

[DataSet3]

		x	y
x	Pearson Correlation	1	.671
	Sig. (2-tailed)		.144
	N	6	6
y	Pearson Correlation	.671	1
	Sig. (2-tailed)	.144	
	N	6	6

7. The weight of a number of copper wires each of length 1m were obtained. These are shown below classified according to die from which they come.

Die No				
I	II	III	IV	V
5	1	5	8	5
4	5	1	5	8
4	5	4	4	5
1	5	5	5	4
4		8	7	
6		1		

8. Are the weights of copper wire for different die are equal?

Solution:

	Row	Column	Value	var	var	var	var	var	var
1	1.00	1.00	5.00						
2	1.00	2.00	1.00						
3	1.00	3.00	5.00						
4	1.00	4.00	8.00						
5	1.00	5.00	5.00						
6	2.00	1.00	4.00						
7	2.00	2.00	5.00						
8	2.00	3.00	1.00						
9	2.00	4.00	5.00						
10	2.00	5.00	8.00						
11	3.00	1.00	4.00						
12	3.00	2.00	5.00						
13	3.00	3.00	4.00						
14	3.00	4.00	4.00						
15	3.00	5.00	5.00						
16	4.00	1.00	1.00						
17	4.00	2.00	5.00						
18	4.00	3.00	5.00						
19	4.00	4.00	5.00						
20	4.00	5.00	4.00						
21	5.00	1.00	4.00						
22	5.00	3.00	8.00						
23	5.00	4.00	7.00						
24	6.00	1.00	6.00						
25	6.00	3.00	1.00						
26									
27									

Oneway

[DataSet4]

ANOVA

		Sum of Squares	df	Mean Square	F	Sig.
Row	Between Groups	4.840	4	1.210	.440	.778
	Within Groups	55.000	20	2.750		
	Total	59.840	24			
Value	Between Groups	16.200	4	4.050	.990	.436
	Within Groups	81.800	20	4.090		
	Total	98.000	24			

9. From the following grouped data find mean, median, quartiles, standard deviation and variance.

Wages (Rs.)	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of persons	2	5	6	22	34	21	9

	 l	 u	 f	 m	var	var	var	var	
1	10.00	20.00	2.00	15.00					
2	20.00	30.00	1.00	25.00					
3	30.00	40.00	3.00	35.00					
4	40.00	50.00	11.00	45.00					
5	50.00	60.00	2.00	55.00					
6	60.00	70.00	5.00	65.00					
7	70.00	80.00	32.00	75.00					
8									
9									
0									
1									
2									
3									
4									
5									
6									
7									
8									

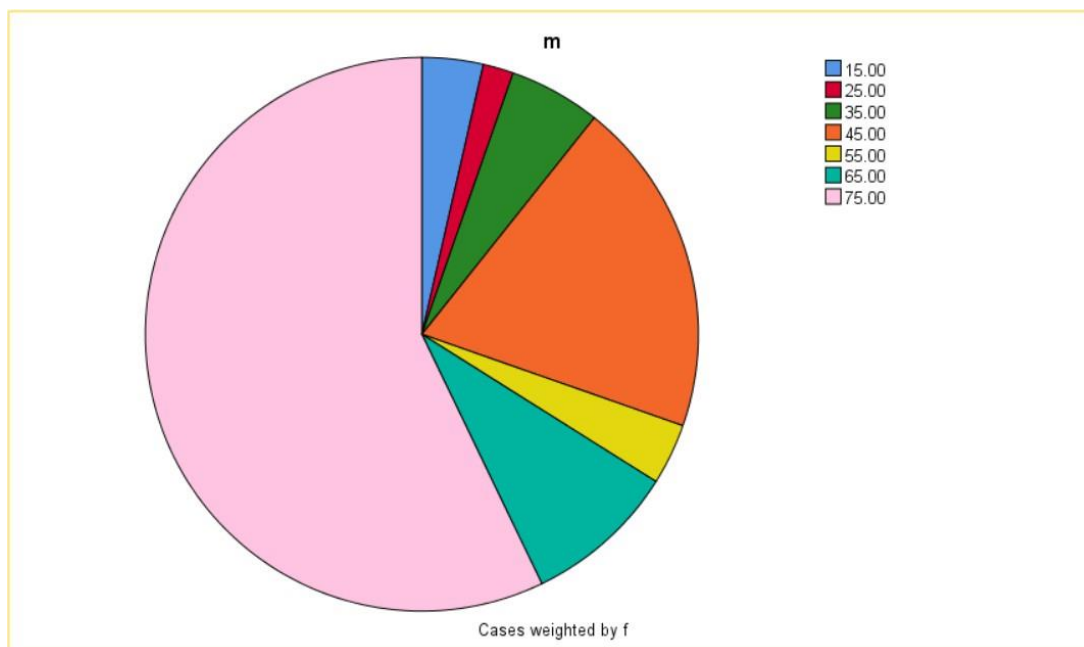


Frequencies

Statistics		
m		
N	Valid	7
	Missing	0
Mean		45.0000
Median		45.0000
Mode		15.00 ^a
Std. Deviation		21.60247
Sum		315.00

a. Multiple modes exist. The smallest value is shown

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	15.00	1	14.3	14.3	14.3
	25.00	1	14.3	14.3	28.6
	35.00	1	14.3	14.3	42.9
	45.00	1	14.3	14.3	57.1
	55.00	1	14.3	14.3	71.4
	65.00	1	14.3	14.3	85.7
	75.00	1	14.3	14.3	100.0
Total		7	100.0	100.0	



FREQUENCIES VARIABLES=m

